# Identifying Seismogenic Anomalies in Satellite Microwave Emissions by Decoupling Environmental Confounders via Physics-Informed Deep Learning

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***Abstract*—Identifying reliable microwave emission anomalies related to impending earthquakes (EQs) remains challenging due to their intricate coupling with surface environmental dynamics. While previous studies rely heavily on historical baselines to remove background emissions, they largely overlook the terrestrial confounding factors inherent within the seismic influence period, compromising the reliability of obtained anomalies. To bridge this gap, we present a novel microwave-physics-informed deep learning (DL) framework that captures the spatiotemporal dependencies of satellite observations across both historical baselines and concurrent influence periods. Distinct from purely data-driven approaches, we embed explicit microwave physical constraints—namely, dual-polarization consistency and gradient monotonicity—directly into the loss function. Validated across multiple EQ cases, our model effectively decouples overwhelming meteorological and surface interferences (e.g., soil moisture variations, precipitation) from the microwave observations, isolating residuals highly associated with targeted EQs. Comparative analysis demonstrates the framework's superiority over traditional techniques, achieving an average 22% increase in contrast-to-noise ratio (CNR) and a 260% enhancement in average anomaly intensity. Ablation studies further confirm that microwave physical constraints significantly enhance reconstruction fidelity, improving the correlation coefficient by 3% and reducing root-mean-square error (RMSE) by 36%. Crucially, validation against ancillary datasets reveals that our framework can account for nonseismic drivers. Consequently, the derived residuals are largely immune to these confounders and inherently seismogenic, substantially reducing the need for *post-hoc* exclusion analyses. This work provides a physically interpretable solution for identifying reliable seismic precursors using multisource satellite remote sensing, bearing profound implications for the future development of an operational, precursor-based EQ forecasting framework.**

***Index Terms*—Deep learning (DL), earthquake (EQ) precursor, microwave brightness temperature (MBT), microwave emission, seismic thermal anomaly.**

## I. Introduction

SATELLITE remote sensing outperforms traditional in situ observations by offering cost-effective, high-resolution synoptic coverage, revolutionizing the monitoring of Earth's thermal state [1], [2], [3], [4], [5]. Compared to the thermal infrared band, passive microwave remote sensing possesses distinct advantages due to its superior penetration through clouds, rain, and aerosols, enabling all-weather observations [6], [7], [8]. More critically, microwave emissions can penetrate surface media to varying depths depending on the frequency and polarization [7], thereby providing a more comprehensive reflection of the Earth's surface thermal state and subsurface thermodynamic variations [9], [10]. These characteristics render satellite microwave emission an effective tool for the real-time, dynamic monitoring of the land surface thermal environment and its subtle variations.

The phenomenon of microwave emission enhancements prior to large earthquakes (EQs) is widely recognized as a potential precursor, and research into its perception and cognition has advanced significantly [11], [12], [13]. The prevailing paradigm involves utilizing historical background, which is based on multiyear averaging [14], spatiotemporal weighting [9], [10], [15], or wavelet decomposition [16], [17], to simulate a natural state assumed to be unaffected by seismic activity. By subtracting this stable background from the observed microwave emission in event year, nonseismic emission interferences are filtered out, thereby retaining precursor signals potentially associated with crustal stress changes and EQ nucleation.

To ascertain that the identified thermal anomalies are physically sound and seismically driven, the lithosphere–coversphere–atmosphere–ionosphere (LCAI) coupling theory is leveraged to cross-validate multiparameter sharing homologous physics [18], [19], [20], [21]. Meanwhile, synergetic analysis using multisource data is also frequently utilized to discriminate the true causes of extracted anomalies. For instance, in the study of the 2017 Iran–Iraq border EQ (Mw 7.3), a striking microwave anomaly strip pointing

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TABLE I
KEY INFORMATION OF THE TYPICAL EQ CASES

| Information | EQ cases | | |
|---|---|---|---|
| | Wenchuan | Nepal | Maduo |
| Event date | May 12, 2008 | April 25, 2015 | May 21, 2021 |
| Magnitude (Mw) | 7.9 | 7.8 | 7.3 |
| Depth (km) | 19 | 8.2 | 10 |
| Geo-location | 31.00°N, 103.32°E | 28.23°N, 84.73°E | 34.59°N, 98.24°E |

toward the epicenter was observed [22]. However, superposed analysis of satellite cloud imagery, precipitation data, and soil moisture revealed this to be a pseudo-anomaly caused by increased surface humidity from a specific rainy cloud. Similarly, in the study of the 2021 Maduo EQ (Mw 7.4) in China, extensive microwave emission enhancement was found northwest of the epicenter [14]. Cross-check attributed this to regional drought conditions, as its spatial morphology and temporal evolution showed no connection to fault zones or the upcoming epicenter.

However, critical limitations persist in current methodologies. Microwave backgrounds constructed solely from historical nonseismic data represent a climatological trend but fail to account for the specific interferences occurring in the EQ-year itself. Consequently, residuals obtained by subtracting the historical backgrounds from EQ-year observations cannot be completely considered as seismic-related, as they are potentially contaminated by concurrent environmental noise. This necessitates post-hoc discrimination using multivariate auxiliary data [14], [22], [23], significantly increasing the workload. Moreover, the availability and reliability of auxiliary data may constrain the operability and robustness of the exclusion analysis.

More fundamentally, previous exclusion analyses attributed these identified thermal anomalies to climatic temperature fluctuations or thick clouds, which addressed the symptoms rather than the root cause. The decisive influence of crustal stress change on the surface microwave dielectric constant was largely overlooked [9], [24], [25]. There remains a significant research gap in how to incorporate nonseismic factors from the EQ-year into the data processing, thereby accurately stripping away the dielectric changes of environmental confounders (e.g., soil moisture change) to reveal the stress-related emissions directly.

In recent years, deep learning (DL) has reached maturity in the geosciences, demonstrating exceptional proficiency in capturing intrinsic coupling among multisource parameters and utilizing time-series data for reliable prediction. Typical applications include precipitation nowcasting [26] and atmospheric temperature modeling [27]. Although DL is frequently criticized for its closed-box nature, tailoring network architectures with domain knowledge and explicit physical constraints—namely, physics-informed DL—can effectively overcome its purely data-driven limitations [28], [29]. Consequently, complex spatiotemporal prediction problems can be robustly solved within a physically consistent framework. In this context, the identification of preseismic thermal anomalies becomes an ideal application, perfectly leveraging the advanced modeling capabilities of such physically constrained architectures.

Surface microwave emission is inextricably linked to land surface temperature (LST) and the dielectric constant [7], the latter of which is dominantly modulated by surface soil moisture (SSM) and vegetation optical depth (VOD) in a complex, hysteretic, and time-variant manner. Leveraging historical nonseismic data to establish the robust coupling relationship between microwave emission and these atmospheric and surface parameters, a synthetic prediction will be generated by feeding the environmental data of the EQ-year into the DL model. The output thus will effectively represent what the microwave emission should be given the current environmental conditions and the absence of EQ impacts. The residuals between the actual observed and predicted values would, therefore, represent the component unexplainable by meteorological factors, i.e., the potential preseismic anomaly.

To bridge these gaps, this study proposes a microwave physics-informed UNet-convolutional long short-term memory (ConvLSTM)-Attention architecture, specifically designed to handle the complex spatiotemporal coupling of multivariate satellite observations. By training the DL model on historical nonseismic periods, we apply it to event years to extract potential seismic signals. The unveiled preseismic microwave anomalies demonstrate the superior effectiveness and stability of the proposed framework, and the embedded microwave physical constraints significantly enhance the reconstruction fidelity of satellite observations. Meanwhile, verifications using multisource auxiliary data confirm that our model effectively mitigates nonseismic interferences, thereby revealing intrinsic microwave emission anomalies associated with seismic activity.

## II. EQ CASES AND DATASETS

### A. EQ Cases

To rigorously evaluate the performance of the proposed framework, we selected three representative high-magnitude EQs (Table I), i.e., the 2008 Wenchuan EQ (Mw7.9), the 2015 Nepal Gorkha EQ (Mw7.8), and the 2021 Maduo EQ (Mw7.3). These cases were chosen as validation benchmarks because they have been extensively documented in prior literature and are widely acknowledged to exhibit significant, characteristic microwave emission anomalies prior to the mainshocks.

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TABLE II
MAJOR PERFORMANCE AND CHARACTERISTIC OF AMSR-E AND AMSR 2 INSTRUMENTS

| Parameters | Performance and Characteristics | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | AMSR-E | | | | | | AMSR-2 | | | | |
| Time of available data | June 2002 – October 2011 | | | | | | June 2012 - present | | | | |
| Frequency (GHz) | 6.9 | 10.65 | 18.7 | 23.8 | 36.5 | 89 | 10.65 | 18.7 | 23.8 | 36.5 | 89 |
| Bandwidth (MHz) | 350 | 100 | 200 | 400 | 1000 | 3000 | 100 | 200 | 400 | 1000 | 3000 |
| Polarization | H, V | | | | | | H, V | | | | |
| NEΔT (K) | <0.34 | <0.70 | <0.70 | <0.60 | <0.70 | <1.20 | <0.70 | <0.70 | <0.60 | <0.70 | <1.20 |
| Swath width (km) | 1450 | | | | | | 1450 | | | | |
| IFOV (km) | 43×75 | 29×51 | 16×27 | 18×32 | 8.2×14 | 3.7×6.5 | 42×24 | 22×14 | 26×15 | 12×7 | 5×3 |
| Dynamic range (K) | 2.7~340 | | | | | | 2.7~340 | | | | |
| Spatial resolution | 50 | 50 | 25 | 25 | 15 | 5 | 50 | 25 | 25 | 15 | 5 |

Geodynamically, all three EQs are governed by the ongoing Indian–Eurasian plate collision, the primary tectonic driver of crustal deformation and stress accumulation across the Qinghai–Tibet Plateau [30]. Despite this shared background, the specific seismogenic structures and rupture mechanisms vary significantly. The Gorkha EQ was a classic interplate thrust event occurring directly along the Main Himalayan Thrust (MHT) [31]. In contrast, the Wenchuan and Maduo EQs represent intracontinental responses to the plateau's eastward extrusion. The former occurred along the Longmenshan fault zone, the steep eastern margin of the Bayan Har block characterized by complex thrust–strike–slip motion [32], while the latter manifested as a distinct left-lateral strike–slip rupture within the interior of the Bayan Har block [33].

### B. Satellite Microwave Data

To ensure continuous and consistent monitoring across the decadal span of the selected EQs (ranging from the Wenchuan EQ in 2008 to the Maduo EQ in 2021), we utilized microwave brightness temperature (MBT) data from the Advanced Microwave Scanning Radiometer (AMSR) series (see Table II). AMSR-E, onboard NASA's Aqua satellite [34], operated from June 2002 to October 2011, providing the necessary historical baseline and coverage for the Wenchuan event. Its enhanced successor, AMSR-2, launched onboard JAXA's GCOM-W1 satellite [35], commenced operations in May 2012, thereby securing data continuity for the Gorkha and Maduo EQs. These datasets have been extensively utilized in prior LCAI coupling studies [21] and have demonstrated superior sensitivity in capturing preseismic thermal anomalies [9], [15], [23].

To deal with the orbital swath gaps in satellite MBT data and account for the large spatial extent of seismogenic zones, a multiday averaging strategy was applied. This avoids spatiotemporal discontinuities in MBT data, which significantly hinder the performance of DL models, and helps to better capture the continuous process of crustal energy accumulation and release prior to large EQs (Section V-C discusses the optimal performance). Furthermore, to address the resolution discrepancies arising from the multiband nature of MBT data, we unified the spatial resolution by resampling all bands to a 0.1° grid for both training and prediction, ensuring scale consistency in the results.

### C. SSM and LST Data

SSM and LST datasets utilized in this study were derived from the NASA Global Land Data Assimilation System Version 2. This product provides a time series of 36 land surface fields at a 3-h temporal resolution, covering the period from January 2000 to the present [36]. We selected the Average Surface Skin Temperature and the 0–10 cm soil moisture content variable to represent LST and SSM conditions, respectively. To align with the satellite overpass time of the MBT observations, we extracted and averaged the data within the 18:00–21:00 time window (UTC). Data preprocessing includes spatially cropping to the target study area extent, applying respective scale factors for physical value retrieval, filtering out outliers, and resampling native 0.25° resolution to a 0.1° grid to ensure spatial consistency with the MBT data.

### D. Vegetation Index Data

Vegetation index data were derived from the MODIS/Terra + Aqua Vegetation Index from the MAIAC Daily L3 Global 0.05Deg CMG V061 product [37]. This dataset provides daily global coverage at a spatial resolution of 0.05° on a climate modeling grid. For this study, daily NDVI layers, which reflect vegetation dynamics in the selected study areas, underwent the same preprocessing pipeline as the SSM and LST data.

## III. METHODOLOGY

In this study, the primary objective is to identify reliable preseismic MBT anomalies by decoupling satellite microwave observations and the physics-constrained background in the event year (Fig. 1). To reconstruct the reliable environmental background, we propose a physics-informed UNet-ConvLSTM-Attention framework (Fig. 2), which integrates a convolutional neural network (CNN), a ConvLSTM network, and a temporal attention mechanism. To ensure physical plausibility, the model is regularized by a hybrid physical consistency loss function incorporating both gradient monotonicity and dual-polarization constraints.

### A. Problem Formulation

We formulate the extraction of seismic MBT anomalies as a spatiotemporal background reconstruction and removal

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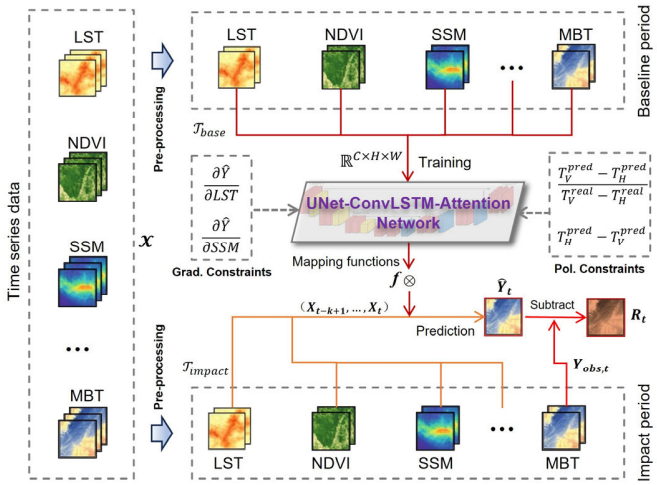



Fig. 1. Workflow of extracting seismic MBT anomalies via removing the microwave-physics-constrained backgrounds. Grad. means gradient, Pol. means polarization.

problem. The satellite-observed MBT, denoted as $Y_{obs}$, is conceptualized as a superposition of a theoretical background component $\hat{Y}$ (governed by surface meteorological factors) and a potential anomaly component $Y_{res}$ (residuals induced by tectonic activity)

$$Y_{obs} = \hat{Y} + Y_{res}. \tag{1}$$

Let the input multivariate time series be denoted as $\boldsymbol{\mathcal{X}} = \{X_t | t = 1, \ldots, T\}$, where $X_t \in \mathbb{R}^{C\times H\times W}$ represents the spatial grid data at time step $t$. $C$ corresponds to the input variables, and $H$, $W$ denote the spatial dimensions. Our goal is to learn a mapping function $f\otimes$ that approximates the background state $\hat{Y}_t$ based on historical and current inputs

$$\hat{Y}_t = f \otimes \left(X_{t-k+1}, \ldots, X_t\right) \tag{2}$$

where $k$ represents the sliding window length. Significant deviations between $Y_{obs,t}$ and $\hat{Y}_t$ are hypothesized to be indicators of seismic MBT anomalies.

## B. Data Preprocessing and Partitioning

First, to address spatiotemporal discontinuities caused by invalid pixels, we applied linear temporal interpolation to restore temporal continuity. For persistent voids where temporal interpolation was insufficient, global mean imputation was employed to ensure the completeness of the input tensor $\boldsymbol{\mathcal{X}}$. Second, to prevent data leakage and preserve the relative magnitude of anomalies, a strict data partitioning strategy was adopted. Impact period ($\mathcal{T}_{impact}$) was experimentally defined as a 31-week window centered on the event week, extending 20 weeks prior and ten weeks postevent. Baseline period ($\mathcal{T}_{base}$) was defined as the historical years excluding $\mathcal{T}_{impact}$.

Data were normalized using Z-Score standardization, where the mean $\mu_{base}$ and standard deviation $\sigma_{base}$ were calculated exclusively from $\mathcal{T}_{base}$

$$X'_t = \frac{X_t - \mu_{base}}{\sigma_{base}} \quad \forall t \in \mathcal{T}_{base} \cup \mathcal{T}_{impact}. \tag{3}$$

## C. Physics-Informed Network Architecture

*1) Spatial Encoder:* As illustrated in Fig. 2, a UNet-style convolutional encoder was utilized to extract the scale-dependent spatial features from each frame $X_t$ and ensures the model captures the regional background emissions before temporal modeling. The use of instance normalization (IN) is critical for stabilizing the distribution of feature maps and preventing internal covariate shifts, especially given the multi-source nature of the input data. Max pooling layers are applied after each block to progressively downsample the feature maps, mapping the input tensor $X_t \in \mathbb{R}^{C\times H\times W}$ to a high-dimensional latent representation $E_t$, effectively compressing spatial redundancy.

*2) Spatiotemporal Memory:* To capture the temporal lag effects of nonseismic variables on surface MBT (e.g., the thermal inertia of soil moisture), we utilized ConvLSTM instead of standard LSTM. ConvLSTM introduces convolutional operations into the state transitions, enabling the model to preserve the spatial topology of the UNet-extracted features while modeling their temporal progression [26]. This design enables the framework to robustly capture the spatiotemporal dependencies of microwave observations across historical baselines and concurrent influence periods. The core transition equations are as follows:

$$i_t = \sigma\left(W_{xi} * E_t + W_{hi} * \mathcal{H}_{t-1} + b_i\right) \tag{4}$$

$$f_t = \sigma\left(W_{xf} * E_t + W_{hf} * \mathcal{H}_{t-1} + b_f\right) \tag{5}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh\left(W_{xc} * E_t + W_{hc} * \mathcal{H}_{t-1} + b_c\right) \tag{6}$$

$$o_t = \sigma\left(W_{xo} * E_t + W_{ho} * \mathcal{H}_{t-1} + b_o\right) \tag{7}$$

$$\mathcal{H}_t = o_t \odot \tanh C_t \tag{8}$$

where $*$ denotes the convolution operator, $\odot$ denotes the Hadamard product. $\mathcal{H}_t$ and $C_t$ represent the hidden state and cell state at time $t$, respectively.

*3) Temporal Attention Mechanism:* Different time steps contribute unequally to the final anomaly identification within a long observational time series. Nonseismic impacts of environmental factors introduce severe but transient noise, whereas seismogenic thermal build-ups present distinct temporal rhythms. A temporal attention mechanism was introduced following the ConvLSTM layer, empowering the model to dynamically assign higher weights to critical phases and suppress nonseismic fluctuations. This mechanism adaptively aggregates temporal features by assigning learnable weights

$$\alpha_t = \text{softmax}\left(W_a \cdot \text{GlobalAvePool}\left(\mathcal{H}_t\right) * E_t + b_a\right) \tag{9}$$

$$\mathcal{Z} = \sum_{t=1}^{k} \alpha_t \mathcal{H}_t \tag{10}$$

where $\mathcal{Z}$ is the fused spatiotemporal context vector, aggregating the most informative features within the sliding window.

*4) Spatial Decoder With Skip Connections:* The decoder aims to reconstruct the high-resolution MBT background from the spatiotemporal context vector $Z$. Crucially, to recover fine-grained spatial details lost during downsampling, a symmetrical skip connection characteristic of U-Net architectures was implemented. These connections concatenate the high-resolution feature maps from the encoder path with the

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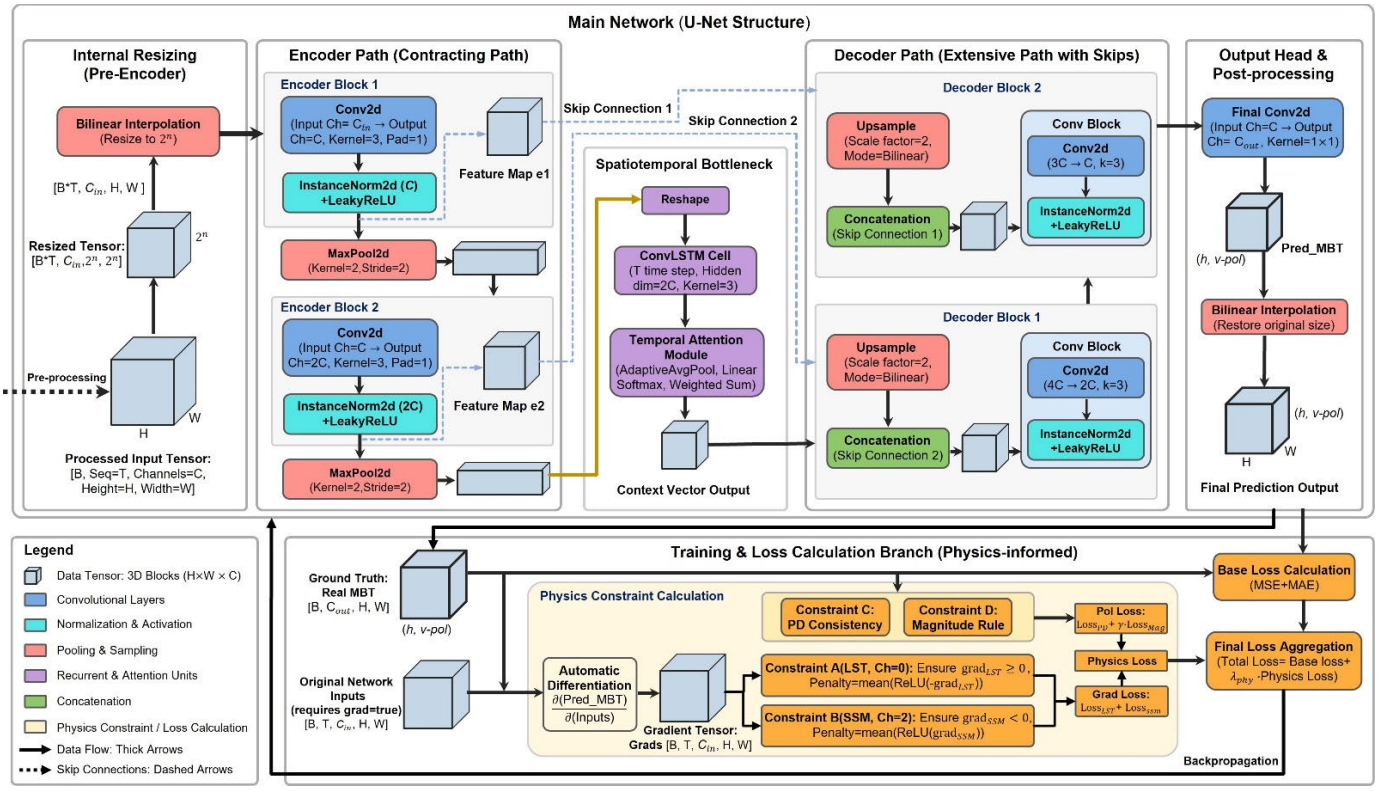



Fig. 2. Detailed architecture of the physics-informed deep neural network developed in this study.

corresponding upsampled feature maps in the decoder path along the channel dimension. This design enables the flow of low-level texture information directly to the output layers, significantly enhancing the precision of the reconstructed background $\hat{Y}$.

### D. Physics-Informed Loss Function

To ensure scientific consistency and generalization capability, we designed a hybrid loss function incorporating explicit microwave physical constraints (see Fig. 2, bottom). Based on microwave emission transfer theory, surface MBT is positively correlated with LST and negatively correlated with dielectric constants (SSM). Meanwhile, governed by Fresnel reflection laws, MBT typically exhibits a distinct polarization pattern where vertical polarization ($T_V$) exceeds horizontal polarization ($T_H$). We leveraged automatic differentiation to compute the gradients of the output with respect to the inputs during training, penalizing predictions that violate these monotonicity and polarization constraints.

The total loss function $\mathcal{L}_{total}$ is defined as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{base} + \lambda_{phy} \cdot \mathcal{L}_{phy} \tag{11}$$

$$\mathcal{L}_{base} = 0.5 \cdot \mathcal{L}_{mse} + 0.5 \cdot \mathcal{L}_{MAE} \tag{12}$$

$$\mathcal{L}_{phy} = \mathcal{L}_{grd} + \zeta \cdot \mathcal{L}_{pol} \tag{13}$$

$$\mathcal{L}_{pol} = \mathcal{L}_{PD} + \mathcal{L}_{Mag} \tag{14}$$

where $\mathcal{L}_{mse}$ focuses on penalizing large errors to ensure overall numerical accuracy, while $\mathcal{L}_{MAE}$ promotes the generation of sharper boundaries and finer textures in the reconstructed image. The hyperparameters $\lambda_{phy}$ serve as balancing coefficients to weigh the contribution of physical constraints against data fidelity, $\zeta$ balances the gradient-based soft constraints ($\mathcal{L}_{grd}$) and the polarization-based hard constraints ($\mathcal{L}_{pol}$).

$\mathcal{L}_{grd}$ refers to constraint A and B in Fig. 2, which penalizes violations of the monotonic relationships between MBT and environmental variables

$$\mathcal{L}_{grd} = \frac{1}{N} \sum_{i} \left[ \mathrm{ReLU}\left( -\frac{\partial \hat{Y}}{\partial LST} \right) + \mathrm{ReLU}\left( \frac{\partial \hat{Y}}{\partial SSM} \right) \right]. \tag{15}$$

The first term penalizes cases where $(\partial \hat{Y}/\partial LST) < 0$ (violating positive correlation with LST), and the second term penalizes cases where $(\partial \hat{Y}/\partial SSM) > 0$ (violating negative correlation with SSM).

$\mathcal{L}_{PD}$ corresponds to constraint C and D (Fig. 2), which enforces the predicted polarization difference ($T_V - T_H$, PD) aligns with the observed intrinsic coupling by employing an L1-norm loss

$$\mathcal{L}_{PD} = \left| \left( T_V^{pred} - T_H^{pred} \right) - \left( T_V^{real} - T_H^{real} \right) \right|. \tag{16}$$

Meanwhile, we apply a ReLU-based penalty ($\mathcal{L}_{Mag}$) to strictly enforce the physical law ($T_V > T_H$)

$$\mathcal{L}_{Mag} = \frac{1}{N} \sum_{i} \left[ \mathrm{ReLU}\left( T_H^{pred} - T_V^{pred} \right) \right]. \tag{17}$$

### E. Anomaly Extraction

The model was trained exclusively on $\mathcal{T}_{base}$ to learn the intrinsic coupling mechanisms of the multivariate system under normal conditions. During $\mathcal{T}_{impact}$, the model parameters

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were frozen and used for forward inference to generate the theoretical MBT backgrounds.

The final seismogenic thermal anomalies were identified by calculating the residual matrix $\mathcal{R}_t$

$$\mathcal{R}_t(h,w) = Y_{obs,t}(h,w) - \hat{Y}_{pred,t}(h,w), \quad t \in \mathcal{T}_{impact}. \tag{18}$$

Given that the model highly captures the coupling relationships inherent in historical observations and involves multivariate inputs during $\mathcal{T}_{impact}$, climatic and hydrological influences are inherently accounted for within the $\hat{Y}$. Consequently, the resulting residuals ($\mathcal{R}_t$) are considered to be less confounded by the environmental drivers and more related to deviations in surface microwave dielectrics caused by crustal stress alterations.

### F. Evaluation Metrics

*1) Contrast-to-Noise Ratio:* The contrast-to-noise ratio (CNR) is a robust metric used to quantify the detectability of an MBT anomaly against its surrounding background. A higher CNR value indicates that the target signal is highly prominent and the background noise is low, making the anomaly easier to detect. It is defined as follows:

$$CNR = \frac{|\mu_A - \mu_B|}{\sigma_B} \tag{19}$$

where $\mu_A$ represents the mean value of the extracted MBT residuals within the defined anomaly region, $\mu_B$ and $\sigma_B$ are the mean value and standard deviation of the background region, respectively. In this study, absolute residuals greater than a global threshold of the mean plus 1.5 times the standard deviation are automatically extracted as target signals, with the rest treated as the background.

*2) Global Moran's I:* Global Moran's I is a fundamental spatial statistic used to measure the degree of spatial autocorrelation within the extracted MBT residuals. The value of Moran's I ranges from −1 to 1. $I > 0$ indicates positive spatial autocorrelation (clustered distribution), $I < 0$ implies negative autocorrelation (dispersed distribution), and $I$ approaching 0 suggests a completely random distribution. The formula is expressed as follows:

$$I = \frac{N}{W} \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} w_{i,j} (x_i - \bar{x}) (x_j - \bar{x})}{\sum_{i=1}^{N} (x_i - \bar{x})^2} \tag{20}$$

where $N$ is the total number of spatial pixels, $x_i$ and $x_j$ are the MBT residuals at locations $i$ and $j$, $\bar{x}$ is the global mean, $w_{i,j}$ represents the spatial weight matrix between pixels $i$ and $j$, and $W$ is the sum of all spatial weights.

*3) Information Entropy:* Information entropy ($H$), derived from Shannon's information theory, quantifies the amount of uncertainty, complexity in MBT residual image. A higher $H$ indicates a highly complex, heterogeneous, and noisy spatial field. A lower $H$ implies a more homogeneous, clean, and orderly spatial distribution (more well-defined and clutter-free MBT anomalies). It is calculated as follows:

$$H = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i) \tag{21}$$

where $n$ is the number of distinct binned residual values in the MBT residual image, and $P(x_i)$ is the probability of occurrence of the value $x_i$.

## IV. Results and Analysis

### A. Time Series of Preseismic MBT Anomalies

Fig. 3(a) depicts the spatiotemporal evolution of MBT residuals surrounding the 2008 Wenchuan EQ (Mw 7.9), derived from the proposed physics-informed framework. The results for Week 0 represent the temporal average of the seven days leading up to and including the mainshock day. Fig. 3(b)–(d) present the time-series profiles of mean MBT residuals for selected regions, smoothed using a Savitzky–Golay filter to reduce noise [38]. The mean ($\mu$) and standard deviation ($\sigma$) in the figures are calculated from the MBT residuals within the specific spatial domain (e.g., Zone A) and temporal window (e.g., from −20 to 10 weeks centered at EQ week) of each respective EQ case.

Spatially, the residual maps reveal that positive values were predominantly concentrated in two distinct zones. Zone A, located approximately 300 km northwest of the epicenter, is tectonically situated at the junction of the Aba, Barkam, and Lixian subblocks. Zone B, positioned southeast of the epicenter, lies within the compression zone between the Longmenshan fault belt [F1 in Fig. 3(a)] and the Longquanshan fault belt [F2 in Fig. 3(a)] in the South China Block. The residual magnitudes in these zones significantly exceeded the background levels of surrounding areas. The deviation-time-space (DTS) criterion [39] suggests that candidate seismogenic anomalies must exceed the predefined threshold and emerge within the impending preseismic time window, located in close proximity to the epicenter or the controlling fault. Accordingly, these sustained residuals can be identified as potential seismic MBT anomalies.

Temporally, the anomaly intensity was already substantial one week prior to the EQ (Week −1). During the EQ week (Week 0), the anomaly in Zone A intensified slightly, whereas Zone B exhibited a dramatic expansion in both intensity and spatial extent. Both anomalies demonstrated rapid attenuation in the week following the mainshock (Week 1). A comparison of polarization channels reveals a high degree of morphological similarity in spatial distribution. However, the intensity of anomalies in vertical polarization (V-pol) was consistently weaker than that in horizontal polarization (H-pol) for both positive and negative residuals. This magnitude disparity suggests that H-pol MBT ($T_H$) possesses superior sensitivity for detecting surface variations, which aligns with the microwave remote sensing physics that V-pol MBT ($T_V$) is less sensitive to surface dielectric changes at specific incidence angles [7].

The temporal evolution of positive MBT anomalies [Fig. 3(b)] elucidates the dynamic accumulation and release of crustal stress. In the 10.65 GHz channel, the MBT anomaly in Zone A began to rise as early as ten weeks prior to the mainshock. In contrast, the anomaly in Zone B exhibited a delayed onset, rising sharply only three weeks before the event. Both zones reached their peak intensity during the EQ week. This spatiotemporal sequence aligns with the geo-

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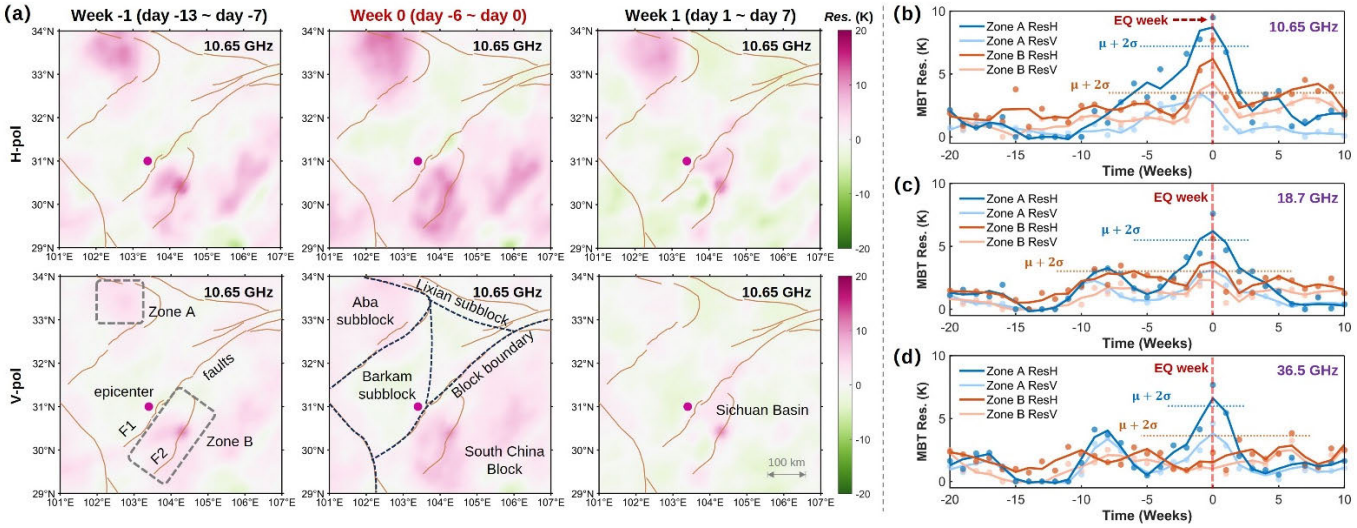



Fig. 3. Spatiotemporal patterns of multifrequency MBT residuals with dual-polarization surrounding the 2008 Wenchuan EQ. (a) Dual-polarized MBT residuals of 10.65 GHz, while (b)–(d) depict the time series of MBT residuals with 10.65, 18.7, and 36.5 GHz. ResH and ResV refer to extracted MBT residuals at H and V polarizations. Timestamps are referenced to the day of the EQ, which is defined as day 0 (last day of week 0). Negative values indicate days prior to the event, while positive values indicate days subsequent to it.

dynamic loading mechanism of the region. The Wenchuan EQ resulted from the persistent eastward extrusion of the Bayan Har block against the rigid South China block. Tectonic stress is hypothesized to have first accumulated and adjusted within the internal subblocks (Zone A: Aba block) due to obstruction [9], followed by stress concentration along the marginal Longmenshan fault zone (Zone B). Following the mainshock, stress release led to a rapid decay of anomalies in both zones.

Regarding frequency dependence, the 18.7 and 36.5 GHz channels exhibited temporal trends similar to those of 10.65 GHz but with notably diminished magnitudes and more pronounced preseismic fluctuations. This indicates that higher-frequency MBT signals possess lower signal-to-noise ratios (SNRs) and stability for anomaly detection. Statistical analysis over the study period (from −20 to +10 weeks) confirms that, for Zones A and B, the 10.65 GHz curves exceeded the significance threshold ($\mu + 2\sigma$) two weeks prior to the EQ, implying a potential forecasting window of 1–2 weeks targeting the two zones.

Conversely, the 18.7 GHz data failed to provide a consistent early warning, and the 36.5 GHz channel showed no significant anomalies throughout the period (particularly in Zone B). These results underscore the superior precursor-detecting capability of 10.65 GHz MBT (especially $T_H$). Physically, this is attributed to the longer wavelength of low-frequency microwaves, which offers stronger penetration capabilities through the atmosphere and surface layers, thereby more accurately reflecting the thermal emission changes induced by subsurface stress.

As illustrated in Fig. 4(a), the extracted MBT residuals for the 2015 Nepal EQ reveal two distinct regions of interest. Zone C is characterized by high-magnitude positive residuals, forming a linear belt that parallels the MHT and encompasses the epicenter. Geographically, this zone represents the rugged transition interface between the Gangetic Plains and the High Himalayas, featuring abrupt topographic gradients. Conversely, Zone D is located within the flat terrain of the Gangetic Plains. As the frontal compression zone of the Indian plate's collision with Eurasia, this region is heavily modulated by the Indian monsoon. Consequently, surface microwave emission in Zone D undergoes drastic fluctuations due to intense precipitation and soil moisture variability, often masking tectonic signals [10].

The temporal analysis [Fig. 4(b)] shows that the low-frequency MBT residuals in Zone C exhibited a sharp intensification 1–2 weeks prior to the mainshock, followed by a rapid postseismic decay. This temporal behavior diverges significantly from the long-term accumulation and slow-release pattern observed in the Wenchuan EQ case. We hypothesize that this distinct impulsive pattern is attributable to the region's high altitude and complex surface characteristics, which may impede the steady surface manifestation of deep-seated thermal emission. Unlike the Sichuan Basin, the harsh surface environment in the Himalayas likely creates a higher detection threshold, allowing anomalies to be observed only when the stress accumulation reaches a critical peak immediately before rupture. However, given the immense magnitude of the event (Mw7.8), the sheer intensity of the seismogenic energy ultimately overcame the surface masking effects, allowing the preseismic anomalies to be successfully captured by satellite sensors.

Notably, a secondary fluctuation in MBT residuals was observed in Zone C approximately 2–3 weeks after the mainshock, which temporally coincides with the major Mw7.3 Dolakha aftershock. This correspondence further validates the seismogenic origin of the extracted signals in Zone C. In contrast, Zone D displayed no such coherent patterns, and the meteorological noises were largely inhibited by our method (Section V-E shows detailed discussions). Furthermore, the frequency and polarization analyses [Fig. 4(c) and (d)] reveal trends consistent with those observed in the Wenchuan case, with lower frequencies and $T_H$ demonstrating superior sensitivity and stability in anomaly detection.

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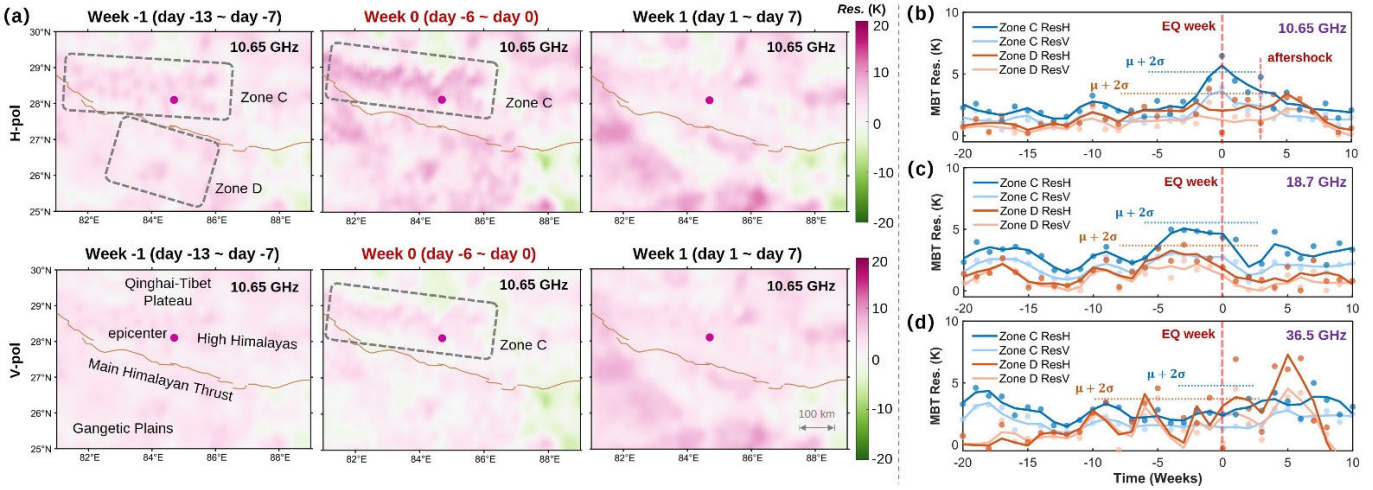



Fig. 4. Time series of multifrequency dual-polarized MBT residuals prior and subsequent to the 2015 Nepal EQ. (a) Shows the spatiotemporal pattern of obtained MBT residuals from 1 week before to 1 week after the EQ, (b)–(d) refer to the temporal changes of abnormnal MBT residuals with 10.65 GHz, (b) 18.7 GHz, (c) and 36.5 GHz, and (d) at dual-polarizations in Zone C and D. The meanings of the text and symbols are the same as in Fig. 3.

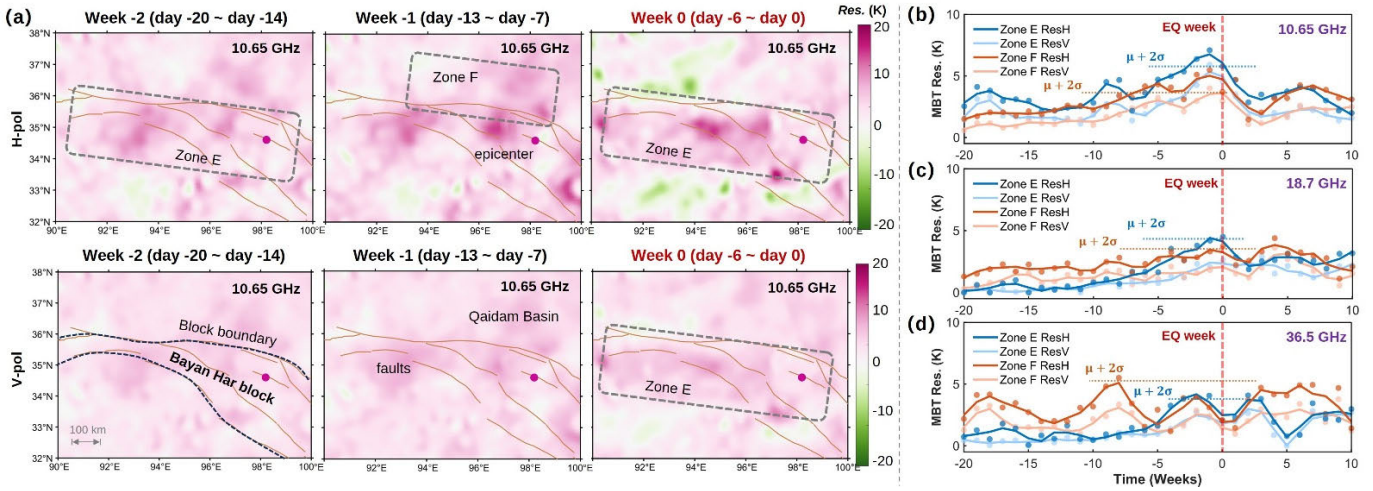



Fig. 5. Time series of multifrequency dual-polarized MBT residuals prior and subsequent to the 2021 Maduo EQ. (a) Shows the spatiotemporal pattern of obtained MBT residuals from 2 weeks before to the coseismic week of the EQ, (b)–(d) refer to the temporal changes of abnormnal MBT residuals with 10.65 GHz, (b) 18.7 GHz, (c) 36.5 GHz, and (d) at dual-polarizations in Zone E and F. The meanings of the text and symbols are the same as in Fig. 3.

For the 2021 Maduo EQ (Mw7.4), the spatial distribution of 10.65 GHz MBT residuals [Fig. 5(a)] reveals two distinct regions exhibiting pronounced positive anomalies. Zone E manifested as a linear belt encompassing the epicenter and extending westward. Tectonically, this region is situated entirely within the interior of the Bayan Har block, a zone characterized by a dense network of faults with predominant strike-slip kinematics. In contrast, Zone F appeared as a patch-like anomaly located to the north of the epicenter, geographically corresponding to the Qaidam Basin. This region is characterized by a hyper-arid desert landscape with sparse vegetation.

Temporally, the thermal enhancement in Zone E exhibited an early onset [Fig. 5(b)–(d)], following a distinct evolutionary pattern with a gradual accumulation prior to the mainshock and a slow attenuation during the postseismic phase. Based on this robust spatiotemporal continuity, this anomaly satisfies the DTS criterion and can be considered seismic-related. Conversely, Zone F only displayed slight EQ-related behaviors in low-frequency results, while results at 36.5 GHz exhibited irregular fluctuations that appeared intermittently throughout both pre and postseismic periods. These variations are likely affected by the drastic SSM variability inherent to the desert surface (a detailed analysis is provided in Section V-E).

### B. Model Performance Validation During Baseline Period

To rigorously evaluate the reconstruction capability of the proposed DL framework, we conducted a comparative analysis between the model-predicted MBT (predicted value) and the actual MBT (observed value) during the baseline period. Taking the 2008 Wenchuan EQ study area as a representative case, Fig. 6(a) presents the spatial distribution of predicted versus observed values for the 10.65 GHz channel (H- and V-pol). Visually, the predicted fields exhibit high fidelity to the observations, with negligible perceptual discrepancies.

Quantitative assessment via scatter plots further confirms this high performance. For the result of $T_H$ at 10.65 GHz, the correlation coefficient ($R$) between predictions and observations reaches 0.95, with a low root-mean-square error (RMSE)

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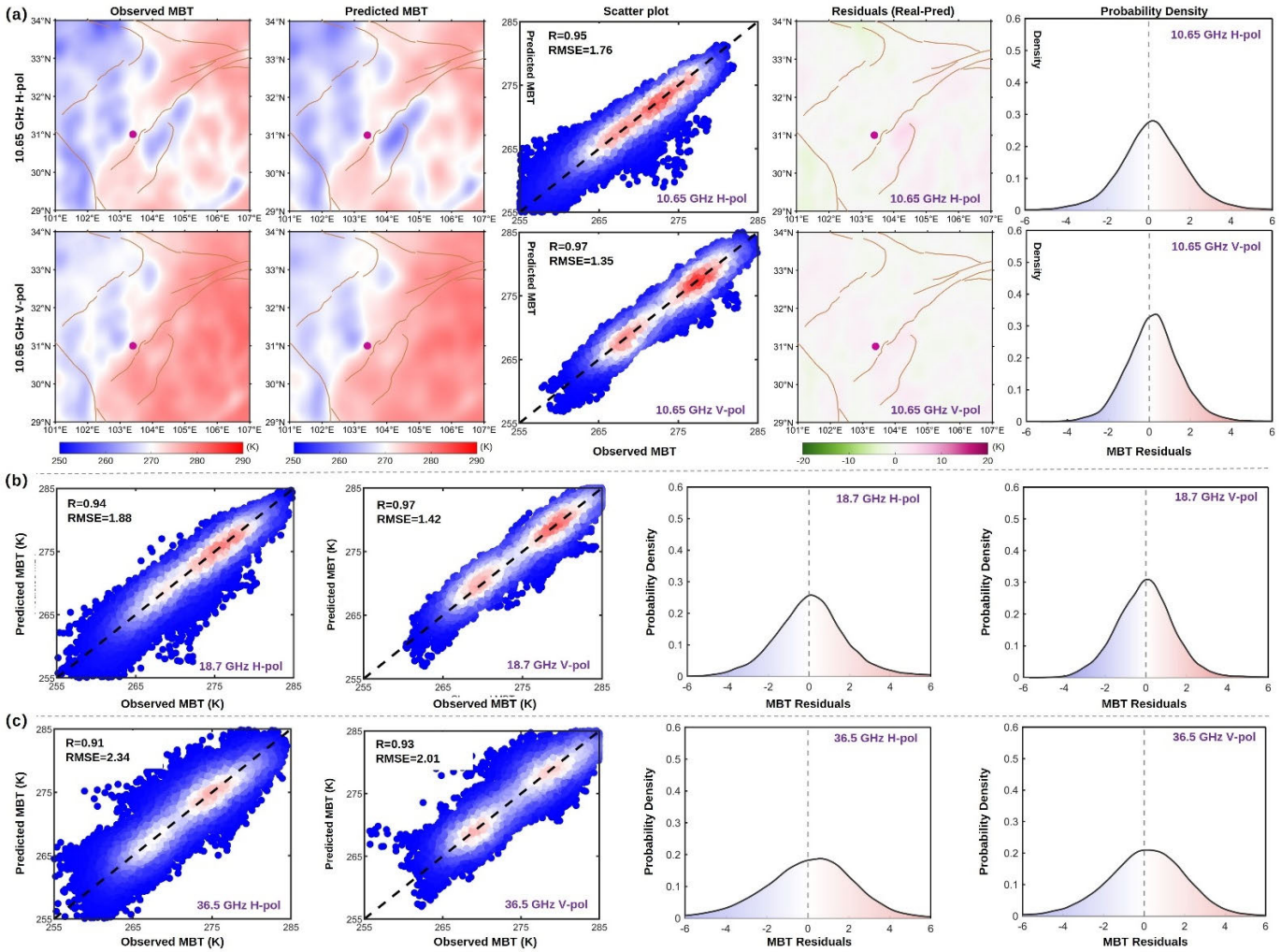



Fig. 6. Validation of the DL model comparing predicted versus observed values. This analysis is based on historical data from the baseline period in the 2008 Wenchuan EQ study area. (a) Spatial comparison and scatter plots of observed and predicted values, along with the spatial distribution and density distribution of residuals for 10.65 GHz (H- and V-pol). (b) and (c) Spatial distribution and density distribution of residuals for 18.7 and 36.5 GHz, respectively.

of 1.76 K. $T_V$ demonstrates even superior accuracy, achieving an $R$ of 0.97 and an RMSE of 1.35 K. Spatially, the residual maps indicate that regions of slight model underestimation (positive residuals) are primarily located on the western Sichuan Basin, while areas of overestimation (negative residuals) are concentrated in the mountainous terrains of the southwest and north. Nevertheless, the residual density plot reveals that the prediction errors are tightly concentrated within the $\pm 2$ K range and exhibit a symmetric distribution centered near zero, indicating no significant systematic bias.

Performance metrics for the 18.7 and 36.5 GHz channels [Fig. 6(b) and (c)] also indicate robust reconstruction capabilities, maintaining the trend where $T_V$ ($R$ = 0.97, 0.93) consistently outperforms $T_H$ ($R$ = 0.94, 0.91). However, statistical accuracy for these higher frequencies is marginally lower compared to the 10.65 GHz results. This frequency dependence can be physically attributed to the input variables of our model that SSM and NDVI exert a dominant influence on the surface dielectric constant. Low-frequency microwave emissions (e.g., 10.65 GHz) are more strongly controlled by these dielectric properties, thereby allowing our model to capture their variability more effectively. In contrast, high-frequency emissions (e.g., 36.5 GHz) have a relatively higher contribution from LST and are more sensitive to atmospheric disturbances.

Crucially, the magnitude of the model's prediction error (~2 K) is significantly smaller than the amplitude of the seismic MBT anomalies of interest (~10 K). This substantial SNR margin confirms that the intrinsic errors of the DL model do not significantly interfere with the extraction of genuine preseismic anomalies.

### C. Comparison of Different Extraction Methods

To validate the superiority of the proposed framework, we benchmarked our results against two representative anomaly extraction methods, i.e., the dynamic monthly mean background (DMMB, [14]) and the spatiotemporally weighted two-step method (STW-TSM, [15]). Fig. 7 illustrates the spatiotemporal evolution of preseismic MBT residuals of the 2008 Wenchuan EQ derived from the three methods. Regarding Zone A (as defined in Fig. 3), while the DMMB

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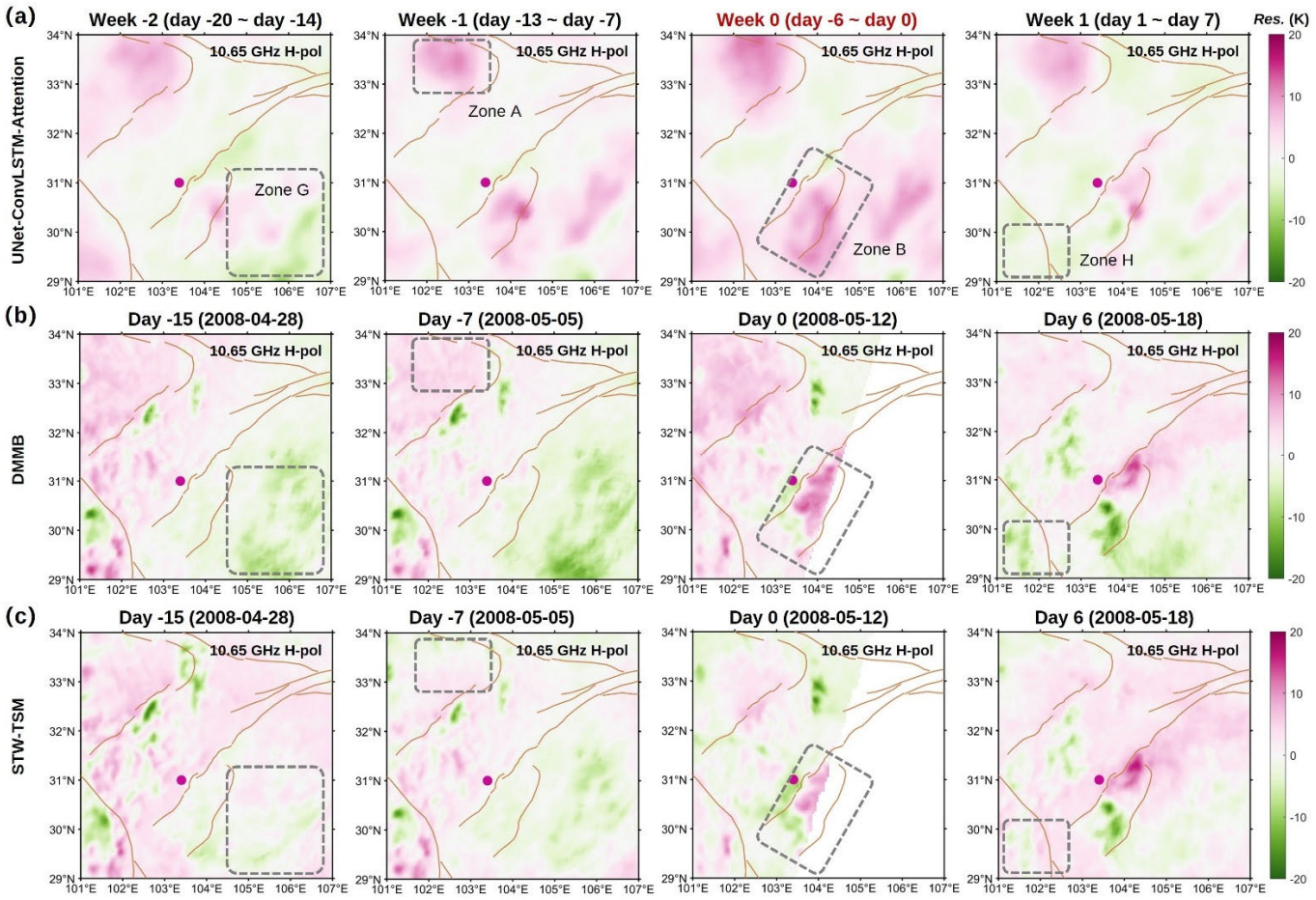



Fig. 7. Comparison of spatiotemporal patterns of MBT residuals extracted by different methods. The data presented correspond to H-polarized 10.65 GHz results surrounding the 2008 Wenchuan EQ. (a) Residuals obtained using the DL-based method proposed in this study. (b) and (c) MBT residuals derived from the DMMB and STW-TSM methods, respectively.

method [Fig. 7(b)] captured anomalous values, they lacked spatial cohesion, presenting a scattered distribution without clear regional clustering. In Zone B, although orbital data gaps partially obscured the DMMB results on the mainshock day (which indirectly validates the necessity of the multiday averaging strategy), a significant anomaly is discernible between the Longmenshan and Longquanshan fault zones. However, crucially, this signal was notably absent in the results from 7 and 15 days prior to the event, indicating a failure to capture the early accumulation process.

For the STW-TSM results [Fig. 7(c)], the regional anomaly in Zone A appeared one week before the EQ week, but its intensity was relatively subdued compared to the DL-based results. Furthermore, the anomaly in Zone B was only identifiable during the mainshock week and the subsequent postseismic week, with no precursory signals detected in earlier temporal windows. In contrast, the DL-based results demonstrated superior performance in both Zones A and B, characterized by higher intensity of regional average anomaly (enhanced by an average of 260% for the two benchmark methods) and more distinct morphological features that align with tectonic structures.

Additionally, in the southeastern Zone G and southwestern Zone H of the study area, the DMMB method exhibited significant negative residuals (likely meteorological artifacts), whereas both STW-TSM and our DL-based method showed minimal deviations. This comparison suggests that our physics-informed framework is more robust in minimizing nonseismic interference while enhancing the visibility of true tectonic signals.

The quantitative evaluation metrics listed in Table III corroborate these visual observations. The DL-based results achieved a significantly higher CNR across all three frequencies and dual polarizations, outperforming the baseline methods by an average of 22%. This indicates that our model effectively suppresses background noise while preserving the amplitude of the seismic anomalies. Furthermore, regarding spatial clustering, our method yielded a satisfactory Global Moran's I (3% better than benchmark methods on average), confirming that the identified anomalies are statistically significant spatial clusters rather than random fluctuations. In terms of Shannon Entropy, no substantial disparities were observed among the three methods (only 1% improvement for the DL-based method).

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TABLE III
PERFORMANCE COMPARISON BETWEEN DIFFERENT METHODS

| Study area | Parameter | Frequency | DMMB | | STW-TSM | | Deep learning | |
|---|---|---|---|---|---|---|---|---|
| | | | H-pol | V-pol | H-pol | V-pol | H-pol | V-pol |
| Wenchuan EQ | CNR ↑ | **10.65 GHz** | 2.07 | 2.14 | 2.25 | 2.19 | **2.76** | **2.58** |
| | | 18.7 GHz | 2.15 | 1.79 | 1.96 | 1.83 | 2.62 | 2.22 |
| | | 36.5 GHz | 2.09 | 2.33 | 2.16 | 2.46 | 2.74 | 2.57 |
| | Global Moran's I ↑ | **10.65 GHz** | 0.95 | 0.93 | 0.95 | 0.94 | **0.97** | **0.96** |
| | | 18.7 GHz | 0.92 | 0.90 | 0.93 | 0.91 | 0.92 | 0.91 |
| | | 36.5 GHz | 0.89 | 0.84 | 0.88 | 0.85 | 0.89 | 0.88 |
| | Entropy ↓ | **10.65 GHz** | 6.76 | **6.57** | 6.58 | 6.64 | **6.54** | 6.83 |
| | | 18.7 GHz | 6.89 | 7.01 | 6.82 | 7.12 | 7.16 | 7.27 |
| | | 36.5 GHz | 6.55 | 6.26 | 6.38 | 6.09 | 5.91 | 5.71 |
| Nepal EQ | CNR ↑ | **10.65 GHz** | **1.93** | 1.87 | 1.85 | 1.74 | 1.91 | **1.85** |
| | | 18.7 GHz | 1.75 | 1.71 | 1.66 | 1.53 | 1.82 | 1.68 |
| | | 36.5 GHz | 1.59 | 1.53 | 1.76 | 1.64 | 1.77 | 1.71 |
| | Global Moran's I ↑ | **10.65 GHz** | 0.91 | 0.90 | 0.89 | 0.85 | **0.92** | **0.93** |
| | | 18.7 GHz | 0.87 | 0.84 | 0.83 | 0.79 | 0.87 | 0.85 |
| | | 36.5 GHz | 0.82 | 0.78 | 0.74 | 0.73 | 0.83 | 0.81 |
| | Entropy ↓ | **10.65 GHz** | 7.16 | 7.05 | 6.88 | **6.62** | 6.68 | **6.57** |
| | | 18.7 GHz | 6.92 | 6.71 | 6.83 | 6.62 | 7.28 | 7.05 |
| | | 36.5 GHz | 6.78 | 6.54 | 6.57 | 6.41 | 6.34 | 6.21 |
| Maduo EQ | CNR ↑ | **10.65 GHz** | 1.77 | 1.64 | 1.75 | **1.69** | **1.82** | 1.68 |
| | | 18.7 GHz | 1.65 | 1.59 | 1.66 | 1.53 | 1.72 | 1.54 |
| | | 36.5 GHz | 1.49 | 1.33 | 1.46 | 1.34 | 1.64 | 1.51 |
| | Global Moran's I ↑ | **10.65 GHz** | 0.85 | 0.82 | 0.86 | 0.84 | **0.87** | **0.85** |
| | | 18.7 GHz | 0.82 | 0.78 | 0.82 | 0.81 | 0.84 | 0.82 |
| | | 36.5 GHz | 0.74 | 0.71 | 0.78 | 0.77 | 0.79 | 0.78 |
| | Entropy ↓ | **10.65 GHz** | 7.56 | **7.32** | 7.68 | 7.67 | **7.54** | 7.33 |
| | | 18.7 GHz | 7.72 | 7.56 | 7.81 | 7.53 | 7.26 | 7.48 |
| | | 36.5 GHz | 7.79 | 7.46 | 7.67 | 7.48 | 6.94 | 6.73 |

Furthermore, we extended our comparative analysis of the three different methods to the Nepal and Maduo EQ study areas. The results consistently demonstrate the superiority of our proposed approach over the other two baseline methods, achieving an average increase of 5% in CNR and 4% in Moran's I, alongside a 3% reduction in information entropy. The comparisons suggest that while the information content is comparable for the three methods, the spatial organization and detectability characteristics of the DL-based method are superior.

## V. DISCUSSIONS

### A. Theoretical Basis and Methodological Justification

In traditional microwave inversion, physical constraints are predominantly applied as static, hard thresholds for quality control or parameter retrieval [40]. For instance, the polarization magnitude rule ($T_V > T_H$) and the monotonic dependence of MBT on SSM, and LST serve as the fundamental basis for the microwave emission transfer equation used in iterative inversion algorithms (such as $\tau-\omega$ model, [41]). For theoretical soil profile analysis, [42] embedded the physical brightness temperature model directly into the loss function. This attempt demonstrated the feasibility of a physics-constrained neural network for multichannel passive microwave analysis.

Although integrating physical principles into remote sensing is well-established, studies enforcing physical consistency constraints on the spatiotemporal dynamics of EQ precursors remain remarkably scarce. Previous methods typically utilize physical laws as static filters or inversion solvers at a single time step. In contrast, our framework embeds these laws directly into the optimization landscape of a DL model via the specially designed loss function. This transforms the physical rules from rigid filters into soft guidance that actively shapes the network's learning trajectory.

Unlike recent developments in physics-informed neural networks (PINNs) which solve partial differential equations for forward problems [28], [43], our approach adapts the PINN paradigm to a data-driven background reconstruction task. By imposing gradient and polarization constraints across the temporal dimension, we force the model to learn the evolutionary rules of the lithosphere–coversphere–atmosphere coupling system, rather than just fitting statistical correlations. This ensures that the reconstructed background respects the intrinsic physical mechanism of dielectric variation even when the input data are perturbed by noise.

The selection of the UNet-ConvLSTM-Attention architecture is not arbitrary but is theoretically grounded in the complex spatiotemporal characteristics of seismic microwave anomalies. Seismic thermal anomalies are often manifested as specific spatial patterns (e.g., linear features along faults or block-wide heating) rather than isolated pixel values. The UNet backbone, with its encoder–decoder structure and skip connections [44], is essential for capturing these multiscale

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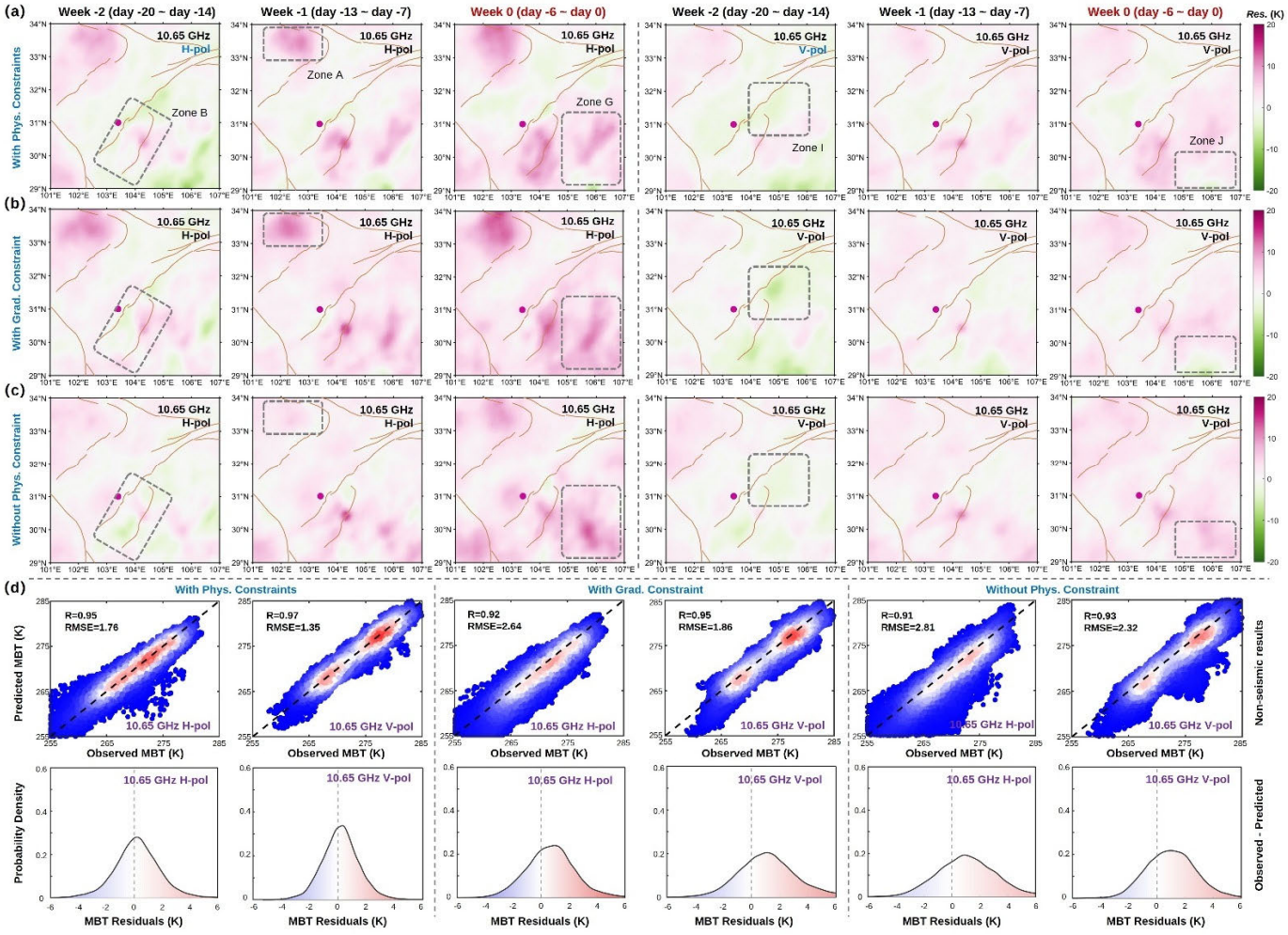



Fig. 8. Results of the ablation study on physical constraints. (a)–(c) Spatiotemporal distributions of MBT residuals prior to the Wenchuan EQ under three conditions: full physical constraints (gradient + polarization constraints), gradient constraint only, and no physical constraints, respectively. (d) Scatter plots comparing predicted versus observed values and residual density distributions, based on baseline data of the study area.

spatial features. It allows the model to preserve high-resolution local textures while modeling the global context of the thermal field, which is critical for identifying faults and their induced boundaries of MBT anomalies.

The interaction between surface microwave emission and environmental factors is inherently hysteretic. For example, the response of surface MBT to soil moisture changes is not instantaneous but depends on the drying/wetting history and soil thermal inertia. Standard CNNs or recurrent neural networks (RNNs) fail to capture this spatial–temporal memory. The ConvLSTM unit addresses this by maintaining a memory state ($C_t$) that encodes historical environmental conditions [26], effectively modeling the time-lagged response of the surface dielectric changes. Meanwhile, the seismogenic process is highly nonlinear, and precursor signals may vary in significance across different nucleation stages. The temporal attention mechanism enables the model to dynamically assign higher weights to critical time steps where the coupling between nonseismic drivers and surface MBT is most informative, thereby filtering out irrelevant temporal noise and enhancing the robustness of the background prediction.

### *B. Rationale for Polarization Constraints*

To validate the necessity of the embedded physical constraints, we conducted an ablation study. We compared the results of the full physics-informed model [Fig. 8(a)] with a variant where the polarization constraints were removed [retaining only gradient constraints, Fig. 8(b)]. Theoretically, three polarization constraints are designed to enforce the fundamental laws of microwave emission transfer. First, the magnitude rule dictates that, based on Fresnel reflection laws at the satellite incidence angle (approx. 55° for AMSR sensors), $T_V$ must physically exceed $T_H$. Second, the sensitivity rule mandates that $T_V$ should be smaller than that in $T_H$. Third, PD consistency requires that the predicted polarization difference ($T_V - T_H$) must align with the observed intrinsic coupling.

The efficacy of these constraints is visually evident in Fig. 8. As highlighted in the Zones I and J ($T_V$ results in Week −2 and Week 0), the full model [Fig. 8(a)] generates residuals with consistent spatial patterns across both polarizations, differing only in magnitude—a behavior consistent with theoretical expectations. In stark contrast, the model

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lacking polarization constraints [Fig. 8(b)] produces physically inconsistent results, exhibiting conflicting signs between H- and V-polarized residuals. This discrepancy arises because, without explicit constraints, the DL network trains the dual-polarization channels independently, neglecting the intrinsic physical coupling between them. Furthermore, the full model successfully extracted significant anomalies in the southeast of the epicenter (Zone B) during Week −1 and Week 0, which were obscured in the unconstrained results.

Quantitative validation [Fig. 8(d)] confirms that removing polarization constraints leads to a marked deterioration in predictive performance for both channels. The correlation coefficient ($R$) decreased by an average of 0.03 (3%), and the RMSE increased by 0.7 K (45%). These findings demonstrate that polarization constraints effectively suppress prediction uncertainty by narrowing the solution space to physically valid states.

The efficacy of the dual-polarization constraint may be limited by surface scattering mechanisms. In densely vegetated or highly rugged regions, dominant canopy volume scattering and complex multiple reflections may distort the intrinsic correlation between H and V polarizations, weakening its ability to constrain deep-seated thermal anomalies.

### C. Rationale for Gradient Constraints

The gradient constraints are grounded in the monotonic relationships governing surface microwave emission, which MBT is positively correlated with LST and negatively correlated with SSM. Therefore, these constraints force the model's partial derivatives to respect these logical gradients. To assess their impact, we further removed the gradient constraints, training the model solely on a mathematical loss function (mse/MAE). The degradation in result quality is substantial [Fig. 8(c)]. Spatially, the model failed to detect the significant regional anomalies in Zones A and B ($T_H$ results in Week −2, Week −1) that were clearly identified by the full model [Fig. 8(a)]. Moreover, the southeastern part of the study area (Zone G) exhibited large-scale nonseismic variations.

This failure can be attributed to the model's inability to distinguish between signal and environmental noise. Zone A is located on the Tibetan Plateau, where LST fluctuates drastically due to complex terrain and weather. Zones B and G (in the Sichuan Basin) are characterized by rapid soil moisture variations. Without the guidance of gradient constraints, the DL model struggles to adapt to these sharp environmental changes, allowing the confounders to mask the subtle preseismic signals. Statistically, removing gradient constraints caused a further decline in model performance. Fig. 8(d) shows that $R$ dropped by an average of 0.02 (2%), and RMSE increased by 0.4 K (26%). This underscores that gradient constraints are critical for decoupling the potential seismic microwave enhancement from the dominant background variations of surface environmental parameters, thereby ensuring the reliability of anomaly detection.

The gradient constraint assumes LST and SSM dominate MBT variations. This assumption may break down under abrupt environmental shifts, such as plateau freezing/thawing cycles which massively alter the dielectric constant or rapid biomass changes which modify canopy optical depth, both of which may affect MBT independently and violate the established gradient relationship.

### D. Influence of Multiday Averaging Strategy

As outlined in the Methodology, we implemented a multiday temporal averaging strategy to preprocess the input satellite data. This approach serves a dual purpose: it mitigates spatiotemporal discontinuities inherent in orbital coverage and, more critically, captures the cumulative thermal integration of seismogenic stress, thereby enhancing resilience against short-term environmental noise. In this part, we evaluated the impact of temporal sliding windows (ranging from 1 to 14 days) on model performance. As illustrated in Fig. 9(a) and (b), the reconstruction accuracy for historical MBT exhibits a strong positive correlation with window size. The temporal smoothing effectively suppresses high-frequency, nonstationary fluctuations in the background environment, resulting in a generally monotonic improvement in performance. The correlation coefficient ($R$) steadily increases while the error (RMSE) progressively declines.

Crucially, however, the sensitivity of precursor detection does not follow this linear trend. As shown in Fig. 9(c), a quantitative analysis of the Wenchuan EQ (specifically comparing anomalies in Zones A and B) reveals a nonmonotonic trajectory in signal intensity. The anomaly contrast initially amplifies as nonseismic noise is filtered, peaking at the seven-day window, before diminishing at longer timescales likely due to the over-smoothing of transient seismic signals. Balancing reconstruction accuracy with anomaly perceptibility, the seven-day composite emerges as the optimal configuration. It achieves the second-lowest RMSE and the third-highest $R$ among all tested intervals, while simultaneously maximizing the contrast of the seismogenic footprint.

### E. Sensitivity Analysis for the Loss Function Weights

To evaluate the impact of the loss function coefficients on the model's predictive performance, we employed a controlled variable approach. The dataset utilized here is identical to that of the study area presented in Fig. 8. To ensure a rigorous comparison, we maintained strictly identical initialization parameters, data partitioning, and training epochs across all coefficient variations. The primary results in this study are derived based on the optimal configuration of $\lambda_{phy} = 0.01$ and $\zeta = 1$, which naturally serve as our baseline for this sensitivity analysis.

By fixing $\zeta = 1$, we first evaluated $\lambda_{phy}$ across a monotonically increasing range (from 0 to 0.5, with an interval of 0.01), with the results depicted in Fig. 10(a) and (c). It is evident that the introduction of physical constraints ($\lambda_{phy} > 0$) leads to a substantial improvement in the correlation between predictions and actual observations, alongside a significant reduction in prediction errors. However, as $\lambda_{phy}$ continues to increase, the physical regularization during training further intensifies, and the model's performance exhibits a fluctuating but downward trend. This indicates that excessive physical regularization may

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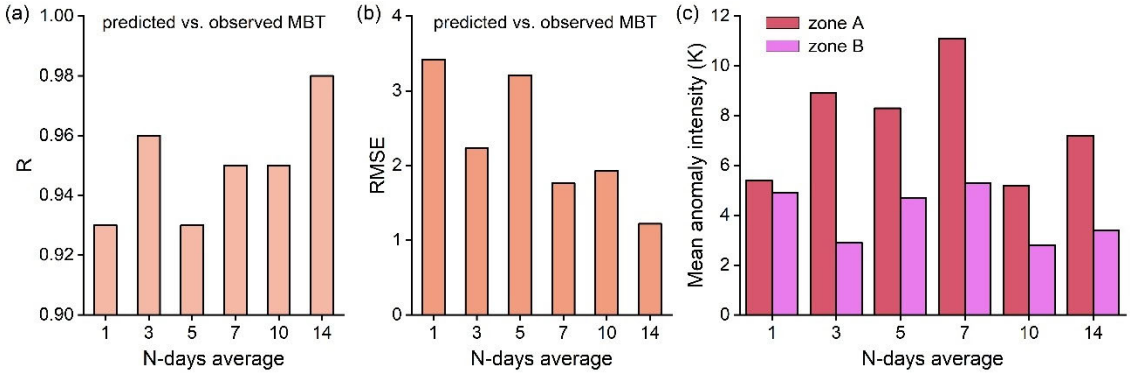



Fig. 9. Statistic results of model performance and derived mean anomaly intensity of different multidays average strategies. (a) and (b) shows the correlation (R) and deviation (RMSE) between the observed and the predicted MBT under different averaging strategies, and (c) shows the different amplitudes of mean MBT a nomalies in Zone A and B.

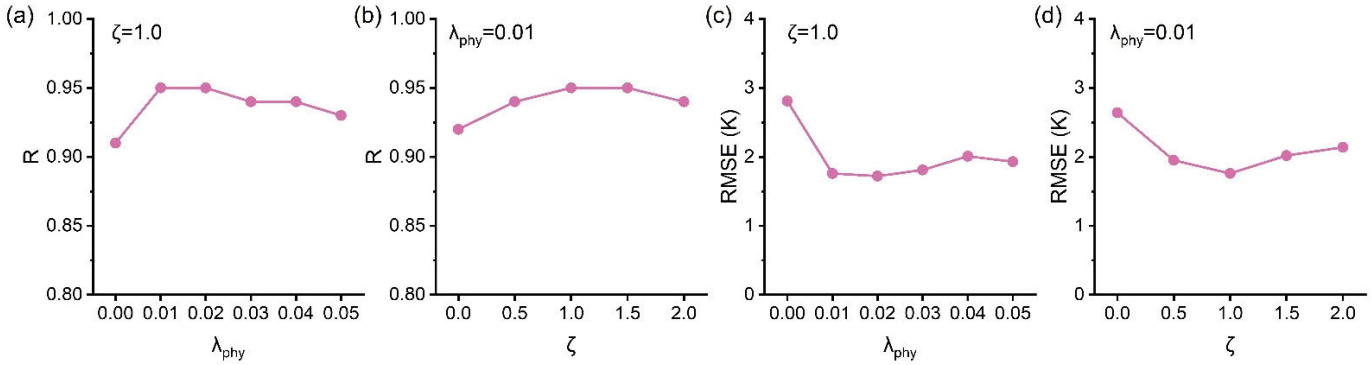



Fig. 10. Sensitivity test for the loss function weights. (a) and (c) represent the R values and RMSE for different $\lambda$ pairs, respectively, when $\zeta$ is held constant, comparing the model's predictions with the actual data, while (b) and (d) show the results when $\zeta$ is the only variable changed, with $\lambda$ held constant.

trigger loss term competition. When $\lambda_{phy}$ becomes disproportionately large, the network may over-prioritize satisfying the simplified physical assumptions, ultimately leading to suboptimal data reconstruction.

Subsequently, we fixed $\lambda_{phy} = 0.01$ and varied $\zeta$ across an increasing set of values (from 0 to 2, with an interval of 0.5), as shown in Fig. 10(b) and (d). When $\zeta = 0$, the polarization constraint is deactivated, leaving only the gradient constraint active. As $\zeta$ increases, elevating the strength of the polarization constraint relative to the gradient constraint, the model's performance initially shows a gradual improvement before eventually declining. This occurs because an optimal balance between the two physical regularizations is essential. Initially, introducing the polarization constraint provides crucial structural decoupling that the gradient constraint alone cannot resolve. However, if $\zeta$ becomes excessively large, the polarization constraint overwhelmingly dominates the regularization process, suppressing the critical spatiotemporal evolutionary relationships captured by the gradient constraint and disrupting their synergistic effect.

### F. Intrinsic Nature of the Extracted Residuals

According to the microwave emission transfer theory, the observed $T_b$ is a product of the surface physical temperature ($T_p$) and emissivity ($\varepsilon$). While $T_p$ contributes linearly, emissivity, which is governed primarily by the surface dielectrics, exerts a more dominant nonlinear influence on the total emission budget. Environmental fluctuations, such as seasonal precipitation or extreme drought, can induce drastic shifts in the dielectric constant, causing $T_b$ variations on the order of tens of Kelvins. These high-amplitude environmental noises can easily mask the subtle seismogenic thermal anomalies (~10 K). Previous studies, attempting to reconstruct the counterfactual nonseismic state, often relied heavily on historical trend analysis [14], [17]. Some approaches incorporated atmospheric removal in the event year's data [15]; they underestimated the decisive impact of surface dielectric variability.

The limitation of traditional methods is evident in the 2015 Nepal EQ's analysis. Significant negative residuals appeared in Zone D (south of the epicenter), which was previously reported to exhibit a seismic-like pattern of preseismic enhancement and postseismic decay [10]. However, auxiliary analysis confirmed it was caused by increased soil humidity, rather than EQ activity. In contrast, our DL framework inherently addresses this by incorporating both the target date's environmental parameters and the antecedent temporal sequences (previous days' data) into the model input. This allows the DL network to learn the localized response of $T_b$ to environmental drivers. As shown in Fig. 11(a), DMMB-based results reflect the precipitation-related moisture response

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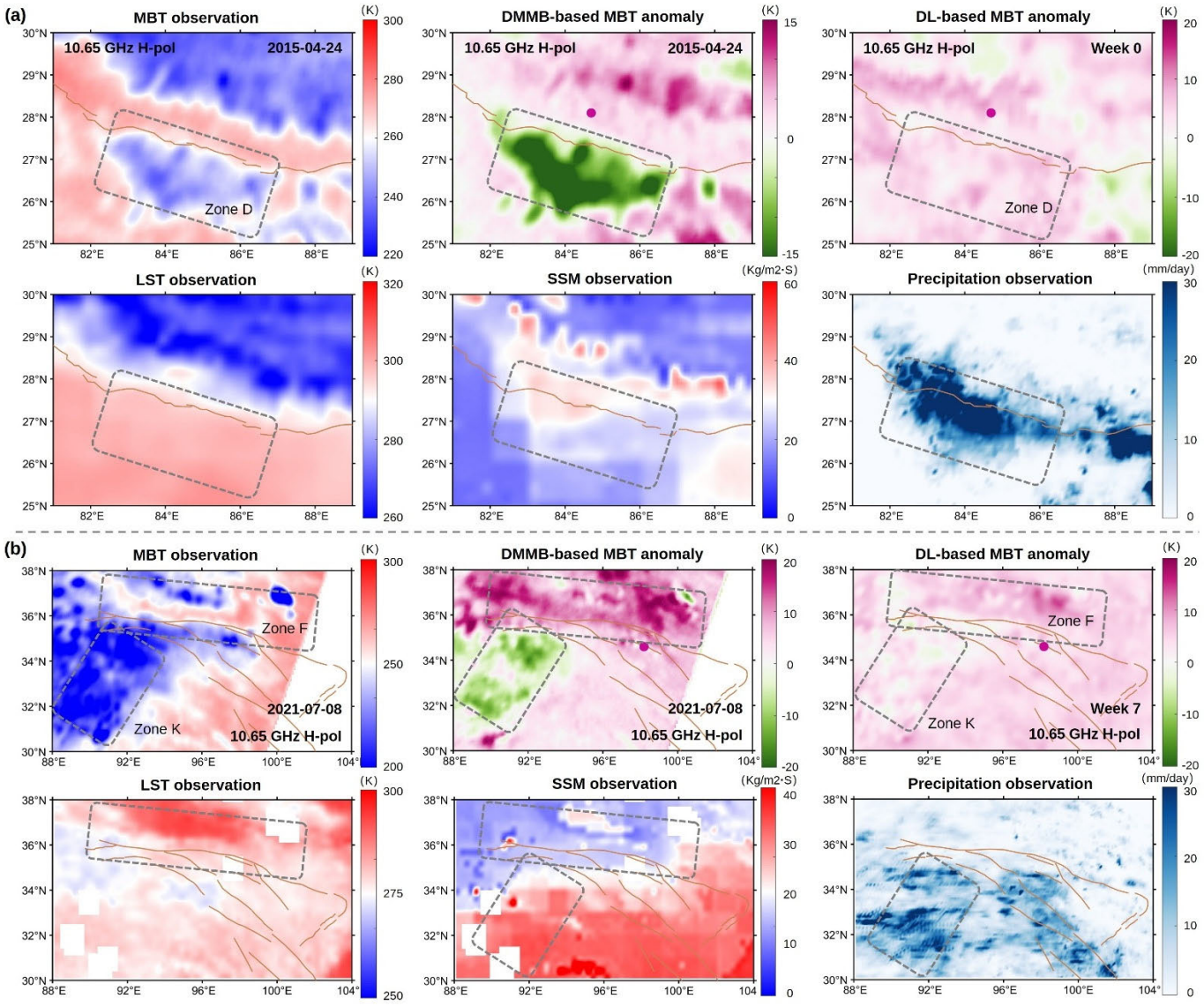



Fig. 11. Comparative validation of the DL-based method in suppressing nonseismic surface disturbances using multiparameter auxiliary data. (a) and (b) Related to the 2015 Nepal EQ and 2021 Maduo EQ, respectively.

in Zone D, but the DL-based method migrates the nonseismic noise without requiring manual intervention.

A similar validation was presented for the 2021 Maduo EQ [Fig. 11(b)]. Results based on the traditional method showed confounding artifacts: Zone F (north of the epicenter) displayed a postseismic MBT rise due to a sudden local drought (dielectric decrease), while Zone K (west of the Bayan Har block) showed an MBT drop linked to rainfall events (dielectric increase). Our physics-informed model successfully captured the rapid environmental shifts within the predicted background. As a result, the derived anomalies by our DL-based method do not exhibit these remarkable nonseismic noises, demonstrating the model's capability to decouple environmental forcing from tectonic signals.

These comparative analyses underscore a fundamental advantage of our approach: it effectively reduces the reliance on laborious post-hoc discrimination using auxiliary data, thereby achieving a *one-step* extraction of seismic anomalies. This not only reduces the workload for analysts but also minimizes the risk of human error in distinguishing between environmental noise and precursor signals. In this study, we use the term *intrinsic*, which denotes that the extracted MBT anomalies are highly associated with crustal stress alteration as well as the consequent changes in surface dielectrics, rather than being artifacts of meteorological or hydrological events.

### G. Implications for EQ Forecasting

By integrating both the concurrent observational input and the antecedent temporal context (spanning the preceding days to weeks), our proposed DL architecture fundamentally enhances the robustness of MBT background reconstruction. Physically, this temporal integration is indispensable because the surface microwave emission state is not a Markovian process governed solely by current conditions; rather, it exhibits significant thermal inertia and hydrological hysteresis. By

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encoding this system memory, our model effectively learns the normal evolutionary trajectory of the surface thermal field under nonseismic conditions. This ensures that the generated background is robust against transient environmental noise.

Crucially, because the framework models the undisturbed baseline, its background reconstruction capability is intrinsically magnitude-agnostic. The generalizability across different EQ magnitudes ($M_w$) depends on the SNR of the isolated residuals. Moderate-to-large events ($M_w \geq 6.0$) typically generate high-amplitude, spatially coherent thermal emissions that significantly exceed the background uncertainty, whereas smaller events ($M_w < 5.0$) may produce weaker signals near the noise floor.

Furthermore, regarding generalizability across diverse tectonic settings, the three seismically active zones in this study represent critical high-stress regions with distinct fault geometries (e.g., strike-slip versus thrust faulting) and geological conditions. The success of our retrospective validation highlights the necessity and efficacy of region-adaptive DL modeling. Unlike rigid global models, our approach leverages historical and concurrent data specific to each region, optimizing parameters to capture local lithological, tectonic, and meteorological dynamics. This regional adaptation strategy provides a flexible pipeline that can be easily recalibrated for other fault systems worldwide, provided sufficient historical satellite archives exist.

Upon detecting a statistically significant MBT anomaly, the potential workflow will transition to a multiparametric cross-validation phase. This procedure involves corroborating the MBT signal with other satellite-derived precursors (e.g., OLR, ionospheric TEC, [45]) and, crucially, seeking verification from ground-based stress/strain observations. This space-ground collaborative validation framework is supposed to significantly reduce false alarm rates, moving us closer to the goal of reliable short-term EQ forecasting.

### H. Shortcomings and Future Improvements

First, a temporal disparity exists between the instantaneous satellite MBT observations and the averaged environmental inputs (3-h LST, daily SSM). The reconstruction accuracy is also intrinsically limited by the uncertainty of the satellite and reanalysis products. Although we selected rigorously validated datasets to ensure baseline reliability, minor alignment errors remain. Future iterations will incorporate higher temporal resolution datasets (e.g., hourly ERA5-Land products) and employ spatiotemporal fusion techniques to minimize the synchronization errors.

Second, the physics-informed DL framework incurs a higher computational overhead compared to traditional statistical baselines. While we optimized efficiency by identifying the data saturation point for training, the cost remains a constraint for large-scale deployment. Future research will explore model lightweighting techniques and transfer learning strategies to accelerate convergence and reduce computational demands for global-scale monitoring.

Third, the current 0.1° spatial resolution is sufficient for regional block analysis but too coarse to resolve fine-scale precursors along narrow fault zones. Super-resolution algorithms guided by high-precision topography might be further employed to downscale the detected anomalies from the block scale to the fault scale, thereby enhancing the spatial precision of seismic hazard assessment.

## VI. Conclusion

To unveil reliable preseismic perturbations on surface MBT from complex environments, traditional methods rely on constructing a historical background and subtracting it from the EQ-year data. However, this paradigm inherently fails to eliminate transient nonseismic interferences that occur specifically within the EQ year. To address this, we proposed a microwave-physics-informed DL framework designed to reconstruct a reliable nonseismic background and thereby eliminate both long-term background trends and short-term nontectonic disturbances.

Comparisons with traditional baseline methods across multiple EQs demonstrate the superior efficacy of our method. Our proposed framework extracts MBT anomalies with more distinct morphological features aligned with tectonic structures and epicenters. Quantitatively, the amplitude of the extracted MBT anomalies is enhanced by approximately 260%, and the CNR outperforms baseline methods by an average of 22% across multifrequency and multipolarization results.

Furthermore, ablation studies underscore the necessity of the embedded physical constraints. The integration of polarization and gradient constraints ensures that the reconstructed background maintains strict physical consistency, particularly in polarization coupling. Compared to unconstrained models, the physics-informed model significantly improves reconstruction fidelity, increasing the correlation coefficient ($R$) by 3% and reducing the RMSE by 36%.

Crucially, auxiliary analyses utilizing multisource data confirm the model's robustness in suppressing environmental noise. The obtained MBT residuals are less confounded by meteorological artifacts, suggesting that the extracted anomalies potentially represent intrinsic seismogenic signals. This capability significantly reduces the dependence on laborious post-hoc discrimination, establishing a *one-step* extraction paradigm. This advancement offers a robust and efficient pathway for identifying reliable seismic thermal precursors, which is of great importance for operational EQ forecasting in the future.

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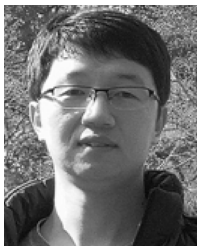
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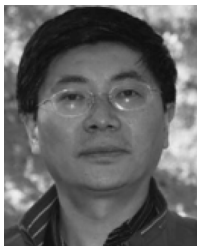
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